CLab 5
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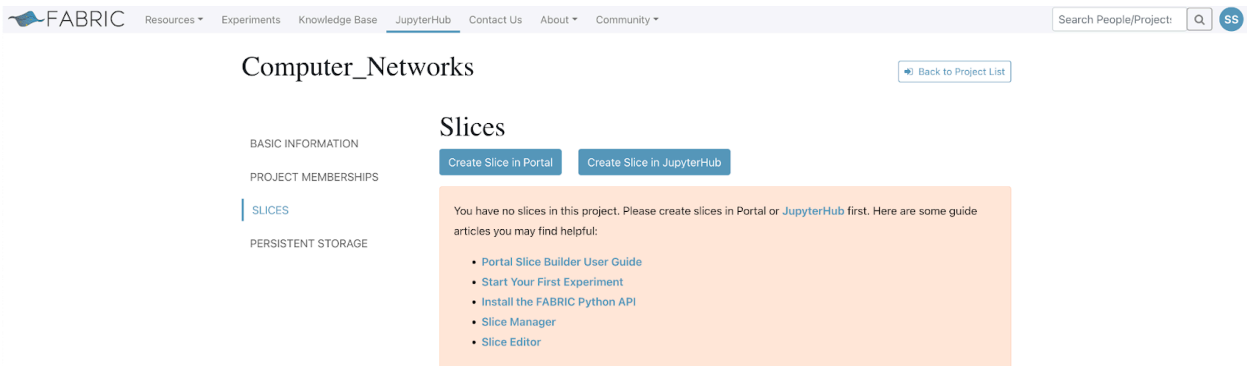
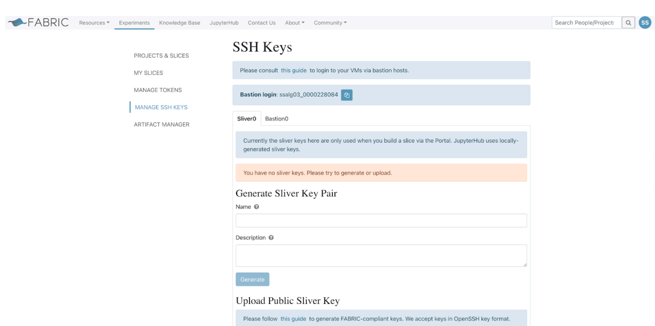
Starting with FABRIC



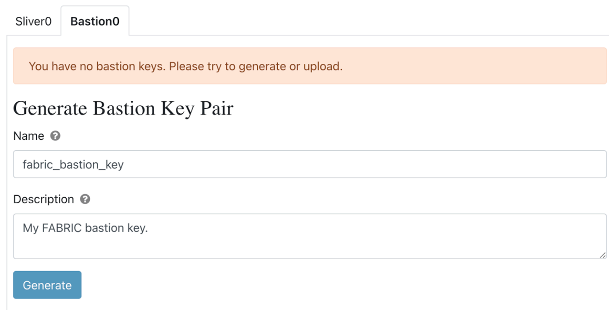
I log into my account and navigate to “Experiments” > “Projects & Slices” > “Computer_Networks” > “Slices”, where I shouldn’t have anything yet. Then, I click on “Create Slice in JupyterHub”.

Setting up and configuring the Jupyter environment

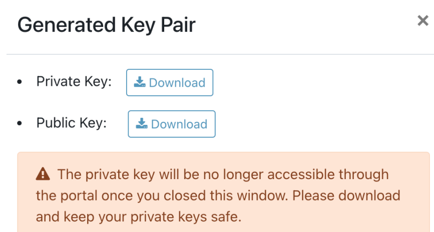
Generating the bastion key



To do this, I click on my profile at the top right of the menu bar, then I navigate to “User Profile” > “My SSH Keys” > “Manage SSH Keys”.



From here, I click on the "Bastion0" tab. Then in the "Generate Bastion Key Pair" section, I enter "fabric_bastion_key" as the name, then I click "Generate".

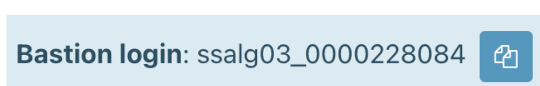


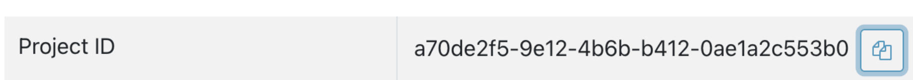
Once generated, I download both keys (private and public).

Uploading the two files to the Jupyter environment

I navigate to the "fabric_config" directory. Then upload both files.

Getting my "bastion username" and "project ID"





Adding my "bastion username" and "project ID"

```
#export FABRIC_PROJECT_ID=a70de2f5-9e12-4b6b-b412-0ae1a2c553b0
#export FABRIC_BASTION_USERNAME=ssalg03_0000228084
```

In the "fabric_rc" file, I include my own information.

Running "configure_and_validate.ipynb"

```
from fabrictestbed_extensions.fablib.fablib import FablibManager as fablib_manager

# Update this line to specify your project id
project_id = "a70de2f5-9e12-4b6b-b412-0ae1a2c553b0"

# Uncomment the line below if using 'FABRIC Tutorials' Project
#project_id="a7818636-1fa1-4e77-bb03-d171598b0862"

fablib = fablib_manager(project_id=project_id)
```

I add my "project ID" then run the cell.

FABlib Config

Credential Manager	cm.fabric-testbed.net
Orchestrator	orchestrator.fabric-testbed.net
Bastion Host	bastion.fabric-testbed.net
Bastion Private Key File	/home/fabric/work/fabric_config/fabric-bastion-key
Slice Private Key File	/home/fabric/work/fabric_config/slice_key
Slice Public Key File	/home/fabric/work/fabric_config/slice_key.pub
Log Level	INFO
Log File	/tmp/fablib/fablib.log
Version	1.7.3
Project ID	a70de2f5-9e12-4b6b-b412-0ae1a2c553b0
Data directory	/tmp/fablib
SSH Command Line	ssh -i {{ _self_.private_ssh_key_file }} -F /home/fabric/work/fabric_config/ssh_config {{ _self_.username }}@{{ _self_.management_ip }}
Core API	uis.fabric-testbed.net
Token File	/home/fabric/.tokens.json
Sites to avoid	
Bastion SSH Config File	/home/fabric/work/fabric_config/ssh_config
Bastion Username	ssalg03_0000228084

Validate the configuration;

- Checks the validity of the bastion keys and regenerates them if they are expired
- Generates Sliver keys if they do not exist already

```
fablib.verify_and_configure()
```

```
User: ssalg03@jaguar.tamu.edu bastion keys do not exist or are expired
Bastion Key saved at location: /home/fabric/work/fabric_config/fabric-bastion-key
Configuration is valid and please save the config!
```

Save the configuration for subsequent use

```
fablib.save_config()
```

I display, validate and save the configuration.

Creating a slice

```python
# Create a slice
slice_name = "StephanieSalgado_Slice1"
site = "EDUKY"
print(site)
nicmodel = "NIC_Basic"
image = "default_ubuntu_20"

cores = 1
ram = 2
disk = 10

try:
    #Create Slice
    slice = fablib.new_slice(name=slice_name)
    A = slice.add_node(name="ND_A", site=site)
    A.set_image('default_ubuntu_20')
    AP1 = A.add_component(model='NIC_Basic', name="APort1").get_interfaces()[0]
    AP2 = A.add_component(model='NIC_Basic', name="APort2").get_interfaces()[0]


    node = slice.add_node(name="Node1")

    slice.submit()
except Exception as e:
    print(f"Slice Failed: {e}")
```

Slice

ID	3e4dc2ab-53a5-4c5b-963d-b8d3eeb5bd60
Name	StephanieSalgado_Slice1
Lease Expiration (UTC)	2024-10-10 17:13:19 +0000
Lease Start (UTC)	2024-10-09 17:13:19 +0000
Project ID	a70de2f5-9e12-4b6b-b412-0ae1a2c553b0
State	Configuring

Slivers

ID	Name	Site	Type	State	Error
7afb6d0a-2640-46c8-9d42-aecda50602b2	ND_A	EDUKY	node	Ticketed	
42191669-98d4-452d-82de-4b5c8ad8f006	Node1	SRI	node	Ticketed	

In “hello_fabric.ipynb”, I run the following cell to create a slice.

Observing the slice's attributes

Show the slice attributes

```
[5]: slice.show();
```

Slice

ID	3e4dc2ab-53a5-4c5b-963d-b8d3eeb5bd60
Name	StephanieSalgado_Slice1
Lease Expiration (UTC)	2024-10-10 17:13:19 +0000
Lease Start (UTC)	2024-10-09 17:13:19 +0000
Project ID	a70de2f5-9e12-4b6b-b412-0ae1a2c553b0
State	StableOK

List the nodes

```
[6]: slice.list_nodes();
```

Nodes

ID	Name	Cores	RAM	Disk	Image	Image Type	Host	Site	Username	Management IP	State	Error	SSH Command
7afb6d0a-2640-46c8-9d42-aecda50602b2	ND_A	2	8	10	default_ubuntu_20	qcow2	eduky-w13.fabric-testbed.net	EDUKY	ubuntu	2610:1e0:1700:206:f816:3eff:fef5:508c	Active		ssh -i /home/fabric/work/fabric_config/slice_key -F /home/fabric/work/fabric_config/ssh_config ubuntu@2610:1e0:1700:206:f816:3eff:fef5:508c
42191669-98d4-452d-82de-4b5c8ad8f006	Node1	2	8	10	default_rocky_8	qcow2	sri-w2.fabric-testbed.net	SRI	rocky	192.5.67.144	Active		ssh -i /home/fabric/work/fabric_config/slice_key -F /home/fabric/work/fabric_config/ssh_config rocky@192.5.67.144

Running the experiment

```python
#node = slice.get_node('Node1')

for node in slice.get_nodes():
    stdout, stderr = node.execute('echo Hello, FABRIC from node `hostname -s`')
```

```
Hello, FABRIC from node NDA
Hello, FABRIC from node Node1
```

Both “ND_A” and the “Node1” created both print their respective messages.

Using the SSH command for Node1

```
fabric@spring:work-16%$ ssh -i /home/fabric/work/fabric_config/slice_key -F /home/fabric/work/fabric_config/ssh_config rocky@2001:400:a100:3080:f816:3eff:fe24:7db3
Warning: Permanently added 'bastion.fabric-testbed.net' (ED25519) to the list of known hosts.
Warning: Permanently added '2001:400:a100:3080:f816:3eff:fe24:7db3' (ED25519) to the list of known hosts.
Activate the web console with: systemctl enable --now cockpit.socket

[rocky@Node1 ~]$
```

I launched a terminal, and entered the SSH command to access “Node1”.

Deleting the slice

```
slice.delete()
```

Slice Name ▲	Slice State	Lease End	
StephanieSalgado_Slice1	Dead	2024-10-10 12:13:19	Slice ID

Finally, after seeing “Hello, FABRIC from node Node1”, I can delete the slice. Notice now, if I were to go to the “Slices” tab, the slice I created above has a status of “Dead” after deleting it.

Summary

In this lab, I used FABRIC to configure a Jupyter environment. I was able to generate a bastion key pair for use in the environment. I used my bastion login username and the project ID in the “fabric_rc” file. Then, I was able to run the configuration and validation notebook provided. After saving the configuration, I moved onto the “hello_fabric.ipynb” file to create a slice. After observing the slice’s attributes, I was able to run the experiment and get the success message: “Hello, FABRIC from node Node1”. Additionally, I used the SSH command provided when listing nodes to access “Node1”. Overall, this was a good introduction to FABRIC. There’s some resemblance between FABRIC and SPHERE, particularly because of the Jupyter environments they both use. This lab was very straight forward.